

Monte Carlo integration

Covered by chapter 6.1-6.3 in the book.

Monte Carlo methods (i.e. statistical simulation) can be used for approximating integrals that are hard/impossible to solve analytically.

This is an alternative to numerical methods for integral approximation. It is particular useful for approximating **high-dimensional integrals**.

Here, we illustrate mainly using 1-d integrals, but keep in mind that we are usually interested in high-dimensional integrals in practice.

Also keep in mind that many quantities relevant in statistics and more generally for characterizing uncertainty may be specified as integrals, e.g. means, variances, probabilities and so on.

Some examples of applications are machine learning, statistics and in high dimensional problems in physics (e.g. in statistical physics and high energy physics.)

Repetition from chap 2:

Recall that if X is a continuous random variable with density $f(x)$ then

$$\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)dx$$

Basic idea

Assume that we want to compute

$$\theta = \int_0^1 g(x)dx$$

If we let $X \sim \text{U}[0, 1]$ this is the same as

$$\theta = \int_0^1 g(x)f_X(x)dx = \mathbb{E}(g(X))$$

Thus if we draw a random sample $X_1, X_2, \dots, X_n$ from $\text{U}[0, 1]$ we can estimate θ by:

$$\hat{\theta}_{SMC} = \frac{1}{n} \sum_{i=1}^n g(X_i) = \overline{g(X)}$$

By the strong law of large numbers $\hat{\theta}_{SMC} \rightarrow \theta$ when $n \rightarrow \infty$.

Notice, no smoothness requirement.

Example: Find $\int_0^1 \cos(x) dx$

See R code: `MC_integration_examples.R`

Easy extension

We now want to compute

$$\theta = \int_a^b g(x) dx$$

This can be written

$$\begin{aligned}\theta &= (b-a) \int_a^b g(x) \frac{1}{b-a} dx \\ &= (b-a) \mathbb{E}(g(X))\end{aligned}$$

and a crude estimate is

$$\hat{\theta}_{CMC} = \frac{b-a}{n} \sum_{i=1}^n g(X_i) = (b-a) \overline{g(X)}$$

where $X_1, X_2, \dots, X_n$ are iid $U[a, b]$.

Example: Find $\int_a^b \cos(x) dx$

See R code: `MC_integration_examples.R`

Example:

Consider the gamma density with shape parameter $\alpha = 4$ and scale parameter $\beta = 0.5$, i.e.

$$f(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} = \frac{8}{3} x^3 e^{-2x}$$

We shall consider two ways of approximating the cdf:

$$F(x) = \int_0^x \frac{8}{3} u^3 e^{-2u} du$$

i) First version is to simulate $U_1, U_2, \dots, U_n$ from the $U[0, x]$ distribution and use the representation:

$$F(x) = x \int_0^x \underbrace{\frac{8}{3} u^3 e^{-2u}}_{g(u)} \underbrace{\frac{1}{x}}_{f(u)} du$$

$$\widehat{F(x)} = \frac{x}{n} \sum_{i=1}^n \frac{8}{3} U_i^3 e^{-2U_i}$$

ii) The second version is to sample from the gamma distribution. Since

$$F(x) = \int_0^x \frac{8}{3} u^3 e^{-2u} du = \int_0^\infty \underbrace{I(u < x)}_{g(u)} \underbrace{\frac{8}{3} u^3 e^{-2u}}_{f(u)} du$$

we can approximate $F(x)$ by simulating $X_1, X_2 \dots X_n$ from the $\text{gamma}(4, 0.5)$ distribution and calculate

$$\widehat{F(x)} = \frac{1}{n} \sum_{i=1}^n I(X_i < x)$$

See R code: `MC_integration_examples.R`

General basic Monte Carlo integration

We want to estimate

$$\theta = \int_A g(x) dx$$

where $g(x)$ can be decomposed into $g(x) = h(x)f(x)$ where $f(x)$ is a probability density on the set A . Then

$$\theta = \int_A h(x)f(x)dx = E(h(X))$$

where X is a random variable with density $f(x)$. Then a general crude Monte Carlo estimate is

$$\hat{\theta}_{GCMC} = \frac{1}{n} \sum_{i=1}^n h(X_i)$$

where $X_1, X_2 \dots X_n$ are iid from f .

Notice, if $g(x) \geq 0$, then one can always chose $f(x) \propto g(x)$, $h(x) \propto 1 \Rightarrow Var(\hat{\theta}) = 0$, but in general obtaining samples from $f(x)$ is roughly as hard as the original problem.

Example:

Find $\int_0^\infty x^5 \exp(-3x) dx$

Notice that $\exp(-3x)$ is an un-normalized exponential distribution density, hence we may rewrite the integral as

$$\int_0^\infty x^5 \exp(-3x) dx = \int_0^\infty \frac{x^5}{3} f(x) dx$$

where

$$f(x) = 3 \exp(-3x)$$

Suggests Monte Carlo estimator

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n \frac{X_i^5}{3}$$

where $X_1, X_2, \dots, X_n \sim \text{iid Exponential with rate parameter } 3$.

See R code: `MC_integration_examples.R`

Extension to higher dimensions

Extension to higher dimensions is immediate *and effective*. E.g in three dimensions

$$\theta = \int_{a_1}^{a_2} \int_{b_1}^{b_2} \int_{c_1}^{c_2} f(x, y, z) dz dy dx$$

can be estimated by

$$\hat{\theta}_{CMC} = \frac{(a_2 - a_1)(b_2 - b_1)(c_2 - c_1)}{n} \sum_{i=1}^n f(X_i, Y_i, Z_i)$$

where $X_i \sim U[a_1, a_2]$, $Y_i \sim U[b_1, b_2]$, $Z_i \sim U[c_1, c_2]$.

The beauty of this is that the computational cost only increases by a factor 3 (not e.g. a factor n^3) - while numerical integration quickly becomes cumbersome in higher dimensions.

Example: Find

$$\int_{-1}^1 \int_2^3 \int_0^1 (xy + yz) dz dy dx$$

See R code: `MC_integration_examples.R`

Example: MC integration on a sphere

MC integration also applies when the domain is not a subset of the real line. Let

$$S^2 = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = 1\}$$

be the unit sphere, which has surface area 4π . Let

$$C = \{\mathbf{x} \in S^2 : x_3 > c\}$$

denote a spherical cap, and let $A = \text{Area}(C)$. If $\mathbf{X}$ is uniformly distributed on S^2 , then

$$P(\mathbf{X} \in C) = \frac{A}{4\pi},$$

and hence

$$A = 4\pi E\{I(\mathbf{X} \in C)\} = 4\pi E\{I(X_3 > c)\}.$$

Thus,

$$\hat{A} = \frac{4\pi}{n} \sum_{i=1}^n I(X_{i3} > c).$$

For $c = 0.5$, the exact area is

$$A = 2\pi(1 - c) = \pi.$$

This illustrates that Monte Carlo integration only requires that we can simulate uniformly from the domain of interest.

See R code: `MC_integration_examples.R`

Variance of MC estimators

For estimating $\int_a^b g(x)dx$ by

$$\hat{\theta}_{CMC} = \frac{b-a}{n} \sum_{i=1}^n g(X_i) = (b-a) \overline{g(X)}$$

where $X_1, \dots, X_n$ are iid $U[a, b]$ we have

$$\begin{aligned} \text{Var}(\hat{\theta}_{CMC}) &= \text{Var} \left(\frac{b-a}{n} \sum_{i=1}^n g(X_i) \right) \\ &= \frac{(b-a)^2}{n^2} \sum_{i=1}^n \text{Var}(g(X_i)) \\ &= \frac{(b-a)^2}{n} \text{Var}(g(X)) = (b-a)^2 \cdot \frac{\sigma_{g(X)}^2}{n} \end{aligned}$$

where we can estimate $\sigma_{g(X)}^2$ by

$$\widehat{\sigma_{g(X)}^2} = \frac{1}{n-1} \sum_{i=1}^n (g(X_i) - \overline{g(X)})^2$$

I.e. the estimated standard error (standard deviation) of $\hat{\theta}_{CMC}$ is

$$\widehat{SD}(\hat{\theta}_{CMC}) = \frac{b-a}{\sqrt{n}} \sqrt{\frac{1}{n-1} \sum_{i=1}^n (g(X_i) - \overline{g(X)})^2}$$

Similarly for estimating $\int_A g(x)dx = \int_A h(x)f(x)dx$ by

$$\hat{\theta}_{GCMC} = \frac{1}{n} \sum_{i=1}^n h(X_i)$$

where $X_1, X_2 \dots X_n$ are iid from f on the set A we get

$$\widehat{SD}(\hat{\theta}_{GCMC}) = \frac{1}{\sqrt{n}} \sqrt{\frac{1}{n-1} \sum_{i=1}^n (h(X_i) - \overline{h(X)})^2}$$

Notice that standard error depends on much variation there is in g or h . If they are very flat, the estimate has small variance.

Efficiency

In general, many Monte Carlo estimates of the same integral are generally possible. One may ask which one is the "best"? Some facts:

- It is generally always possible to find MC estimators with zero or very small variance. However, obtaining draws is very computationally expensive.
- Precision (to be discussed shortly) of MC estimate goes as $O(n^{-\frac{1}{2}})$.
- CPU Cost typically goes as $\text{constant} + O(n)$.

Must take computing time into account when comparing different Monte Carlo estimates.

Rule of thumb;

low dimension: typically OK with crude estimator and large n .

high dimension: generally more refined estimators and small n required.

Asymptotic Normality

Notice that the MC estimators considered here are based on calculating averages, i.e. according to the central limit theorem

$$\frac{\hat{\theta} - \theta}{\sqrt{\text{Var}(\hat{\theta})}} = \frac{\hat{\theta} - \theta}{\text{SD}(\hat{\theta})} \approx \text{N}(0, 1)$$

Thus we can make the same consideration as before (chap 2) that there is approximately a $(1 - \alpha)100\%$ probability that $\hat{\theta}$ falls in

$$\left[\theta \pm z_{\alpha/2} \text{SD}(\hat{\theta}) \right]$$

In all the considered estimators we have that $\text{SD}(\hat{\theta})$ is on the form $\text{SD}(\hat{\theta}) = K/\sqrt{n}$ where K is some constant. Thus we have the same issue as before - precision increases by $\sqrt{n}$.

Example:

Consider the crude MC estimator for which

$$\text{SD}(\hat{\theta}_{CMC}) = (b - a)\sigma_{g(X)}/\sqrt{n}$$

I.e. to have a $(1 - \alpha)100\%$ probability that $\hat{\theta}_{CMC}$ is at most a margin ϵ from θ we need:

$$z_{\alpha/2} \frac{(b - a)\sigma_{g(X)}}{\sqrt{n}} < \epsilon \quad \Rightarrow \quad n > \frac{z_{\alpha/2}^2 (b - a)^2 \sigma_{g(X)}^2}{\epsilon^2}$$

As before $\sigma_{g(X)}^2$ can be estimated by a first simulation and then plugged into the formula.

I.e., we can always increase precision by increasing the number of simulations, but this is a slow increase and the computational costs increases fast - **is there something smarter to be done?**